

MULTI-AGENT SCALING ACROSS DISJUNCTIVE AND COMPENSATORY TASKS

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ABSTRACT

Multi-agent LLM systems are often expected to improve as team size increases, yet the scaling behavior may depend on task structure. *Our central contribution is to introduce Steiner’s taxonomy of group tasks as a framework for analyzing multi-agent LLM scaling* and focusing the analysis on disjunctive and compensatory tasks. We model independently sampled agents as conditionally independent given the item, which yields their large-team limits: plurality voting converges to the model’s modal answer, and averaging converges to the model’s item-level bias. Across selected representative benchmarks, 13 open-weight models, and teams of up to 30 agents, we find qualitatively different scaling behavior. On disjunctive tasks, the probability that at least one agent is correct grows by 5–20 points with team size, but plurality voting over agents that answer directly realises almost none of this potential, as the model predicts to within 0.5 points on average. Multi-round revision raises accuracy considerably, yet the gain is nearly the same with one peer as with 29. In contrast, scaling provides little benefit on Fermi estimation, despite its natural suitability for aggregation: item-level biases shared across the samples of a model account for about 87% of the squared error, so averaging reduces error by only about 6%. Combining model families helps on Fermi estimation but does not surpass the strongest member on disjunctive tasks. These results show that task structure, together with the mechanism combining member outputs, is a fundamental determinant of team scaling. [Open code placeholder]

1 INTRODUCTION

Motivation. With the rapid emergence of Large Language Model (LLM) multi-agent frameworks (Wu et al., 2024; Li et al., 2023; Du et al., 2024), researchers have deployed LLM ensembles and multi-agent debates across a multitude of reasoning and decision-making problems under the implicit assumption that scaling team size N universally enhances system performance. As LLMs are trained on human generated knowledge largely encoded through language, that inherently also captures cultural and behavioral traits, we turn to observing multi-agent behavior in a similar way human group behavior was observed and studied. The ability of a group to outperform its individual members is among the most studied phenomenon in cognitive science, economics, and social psychology. In his classic 1907 experiment, Francis Galton observed that while individual fairgoers made highly scattered guesses regarding the weight of an ox, the median of their estimates fell within 0.8% of the true weight (Galton, 1907; Surowiecki, 2004). Conversely, the foundational Ringelmann effect (Ingham et al., 1974) reveals the opposing tension in collective dynamics: as group size increases, teams frequently suffer from diminishing per-member returns and performance saturation.

In Ivan Steiner’s seminal taxonomy of group processes (Steiner, 1972), resolving whether a collective achieves Galtonian crowd wisdom or succumbs to Ringelmann-style process loss depends on the underlying task structure. The tasks observed by Galton are the so-called **compensatory tasks** Steiner (1972) where group performance is determined by statistical aggregation of continuous estimates. In human crowds, because individual participants possess idiosyncratic cognitive biases and independent heuristic noise, their random errors compensate for one another (Larrick & Soll, 2006; Davis-Stober et al., 2014).

Ringelmann originally observed this performance drop on **additive tasks**, where per-member productivity decayed as group size increased due to coordination losses and social loafing (Latané et al., 1979; Ingham et al., 1974). It was later shown by Kerr & Bruun (1983) that group performance losses are fundamentally mediated by Steiner’s task typology and the perceived dispensability of individual effort. While additive and **conjunctive tasks** frequently trigger free-rider effects or coordination bottlenecks, **compensatory tasks** can circumvent these losses when members provide independent, unbiased estimates that aggregate symmetrically. This established that collective human team scaling laws are fundamentally **task-dependent**. The same observation has been recently made by Kim et al. (2026): LLM agent team performance is task dependent. At the same time, inspired by Ringelmann’s findings, Bertalanic & Fortuna (2026) developed a scaling law for effective team size.

Prior art. Without classifying them into the Steiner taxonomy nor recognizing them explicitly as disjunctive or compensatory tasks, the community has already investigated related multi-agent performance. The rich state of the art on disjunctive tasks represented by benchmarks such as MMLU, GSM8K, MATH500, ARC reports gains from sampling and voting (Wang et al., 2023; junyou li et al., 2024), and from debate (Du et al., 2024) although these gains can saturate or reverse as the number of calls grows (Chen et al., 2024; Bertalanic & Fortuna, 2026). Debate induces conformity (Weng et al., 2025), and correlated errors limit ensembles (Kim et al., 2025). Improvements have also been reported for compensatory tasks represented by Fermi Kalyan et al. (2021); Liu et al. (2025) and guesstimate Chuang et al. (2025) problems. The closest related works are Chen et al. (2024), who explain non-monotone voting by a mixture of easy and hard queries, the basis of our plurality limit, and Bertalanic & Fortuna (2026), who fit a law for the effective team size N_{eff}/N across 44 conditions and find that only heterogeneous teams escape hard ceilings; homogeneous teams also saturate in Yang et al. (2026), and LLM teams fall short of their best member in Pappu et al. (2026).

This work. *Our central contribution is to introduce Steiner’s taxonomy of group tasks as a framework for analyzing multi-agent LLM scaling* enabling the formalization and characterization of multi-agent performance on formal task types:

1. We connect benchmarks extensively used the evaluation of LLM systems to the Steiner taxonomy and mathematically formalise agent behaviour on these tasks, including the large-team limits of independently sampled agents (Section 2).
2. We investigate multi-agent scaling with 13 open-weight models and teams of up to 30 agents, analysing the independence and diversity assumptions (Section 3.2). On disjunctive tasks, $\text{pass}@N$ grows steadily with team size, whereas plurality voting converges to the model’s modal answer, as predicted to within 0.5 points by a conditional-independence model. Revision rounds raise accuracy, but almost equally with one peer or 29 (Section 3.3).
3. We show that unbiasedness fails for compensatory Fermi estimate tasks at the level of individual items: item-level bias accounts for about 87% of the squared log-error, so averaging reduces error by only about 6%; part of the apparent cross-model correlation is due to implausible reference labels (Section 3.4).
4. We show that heterogeneous teams improve substantially on their average member but, on disjunctive tasks, don’t exceed their strongest member; on the compensatory Fermi estimation, a team of five 7B–8B models outperforms its members (Sections 3.3 and 3.4).

2 TAXONOMY AND THEORETICAL SCALING FRAMEWORK

2.1 STEINER’S TASK TAXONOMY AND MODERN LLM BENCHMARKS

We map representative benchmarks used to evaluate LLM-based multi-agent teams onto Steiner’s task types (Steiner, 1972), according to the rule by which a multi-agent system combines member contributions (Table 1); the mapping describes a combination rule rather than an intrinsic property of a benchmark. The remainder of the paper focuses on disjunctive tasks, which depend on the skill of individual members, and compensatory tasks, which depend on error cancellation; we leave the other task types for future work.

Table 1: Categorization of tasks according to the Steiner taxonomy. Disjunctive and compensatory tasks are briefly discussed in this section; with additional details on all tasks in Appendix A.

Task	Combination Operator	Output Domain	Top-Conference Benchmarks (2024–2026)	LLM Aggregation Mechanism
Dis-junctive	$Y = \max_i \{y_i\}$ (or plurality)	Discrete ($\mathcal{Y}$)	MATH-500, GSM8K, MMLU-Pro (Wang et al., 2024), GPQA (Rein et al., 2024), BigCodeBench (Zhuo et al., 2025), ARC	Plurality voting, verifier selection, multi-agent debate
Compensatory	$Y = \left(\prod_{i=1}^n x_i\right)^{\frac{1}{n}}$ (or median)	Continuous ($\mathbb{R}^+, [0, 1]$)	ForecastBench (Karger et al., 2025), Guesstimate (Chuang et al., 2025), RealFP (Kalyan et al., 2021), SynthFP, SciOly	Geometric/arithmetic mean, median, calibrated Brier aggregation
Additive	$Y = \sum_i y_i$ or $\bigcup_i S_i$	Cumulative sets	DAT (Olson et al., 2021), open-ended ideation, parallel instruction synthesis	Deduplicated set union, parallel greedy generation
Con-junctive	$Y = \min_i \{y_i\}$	Boolean / state chain	SWE-bench (Jimenez et al., 2024), TheAgentCompany (Xu et al., 2026), OSWorld (Xie et al., 2024), BrowseComp (Kim et al., 2026)	When modeled as strict sequential pipeline, multi-step dependency verification
Dis-cre-tionary	$Y = f_{\text{group}}(\{y_i\})$ Dynamic	Heterogeneous	AgentBench (Liu et al., 2024), ChatEval (Chan et al., 2024), MoA (Wang et al., 2025)	Meta-agent router, judge arbitration, dynamic role assignment

Disjunctive Tasks in which the outcome is established through $Y = \max_i \{y_i\}$ or plurality/majority/verifier selection and/or debate. The tasks are unitary and optimizing and group success requires that *at least one* agent discovers a valid solution, and the group then recognizes it as such. Examples include mathematical reasoning, multiple-choice knowledge and science questions, and code generation (Table 1), which MASes address through plurality voting (Wang et al., 2023), verifier re-ranking, and debate (Du et al., 2024; Wang et al., 2025).

Compensatory Tasks in which the outcome is determined by $Y = \text{Aggr}(x_1, \dots, x_n)$ via statistical aggregation (geometric mean, arithmetic mean, or median) of continuous numerical estimates ($x_i \in \mathbb{R}^+$) or calibrated probabilities ($p_i \in [0, 1]$). Examples include forecasting (Karger et al., 2025), guesstimation (Chuang et al., 2025), and Fermi estimation (Kalyan et al., 2021; Liu et al., 2025).

2.2 MODELING DISJUNCTIVE TASKS THROUGH VOTING

2.2.1 THE PURE DISJUNCTIVE CASE

In Steiner’s classical disjunctive framework (Steiner, 1972), a task is disjunctive if group success requires only that *at least one* team member generates the correct solution $y^* \in \mathcal{Y}$ and that the group recognises it. Let $p \in (0, 1)$ denote the probability that an individual agent independently produces the correct answer. For independent agents and an ideal verifier, the probability of collective success is $P_{\text{disj}}(N) = 1 - (1 - p)^N$, so the collective failure probability decays exponentially with team size N : $P(\text{failure}) = (1 - p)^N \rightarrow 0$ when $N \rightarrow \infty$. We refer to this, estimated empirically, as the proportion of items on which at least one agent is correct, as $\text{pass}@N$.

2.2.2 MAJORITY VOTING AND THE CONDORCET JURY THEOREM

In the absence of an oracle verifier, multi-agent frameworks aggregate discrete candidate outputs through majority or plurality voting (Wang et al., 2023; Du et al., 2024). For a binary decision between alternative solutions, let each agent cast an independent vote $V_i \sim \text{Bernoulli}(p)$ for the correct solution y^* , where $p > \frac{1}{2}$. The majority vote decision $\hat{y}_{\text{maj}} = \sum_{i=1}^n V_i > \frac{n}{2}$ succeeds with probability $\sum_{k > N/2} \binom{N}{k} p^k (1 - p)^{N-k}$. By Hoeffding’s inequality under Bernoulli random variables where $a_i = 0, b_i = 1$ and deviation $t = p - \frac{1}{2}$,

the probability that the majority vote errors is strictly upper-bounded by:

$$P(\hat{y}_{\text{maj}} \neq y^*) \leq \exp\left(-2n(p - 1/2)^2\right) \quad (1)$$

Under uncorrelated agent assumption, Eq. 1 shows that for agent accuracies strictly greater than random guessing ($p > 0.5$), voting ensembles are bounded by exponential error suppression as team size N increases.

2.3 MODELING COMPENSATORY TASKS THROUGH GEOMETRIC AGGREGATION

Next, we contrast the disjunctive voting paradigm with the classic statistical model of group judgment on **compensatory tasks** (Galton, 1907; Surowiecki, 2004; Steiner, 1972). Because continuous estimation tasks (e.g., Fermi problems) span many orders of magnitude, errors are evaluated in logarithmic space (Kalyan et al., 2021). Let $g \in \mathbb{R}^+$ be the positive ground truth, and $x_i \in \mathbb{R}^+$ be agent i 's estimate for $i \in \{1, \dots, N\}$. The signed log-error is $e_i = \log_{10}(x_i/g)$. The team estimate $\bar{X}_N$ is formed via geometric mean aggregation, corresponding to arithmetic mean in log-error space:

$$\bar{e}_N = \log_{10}\left(\frac{\bar{X}_N}{g}\right) = \frac{1}{N} \sum_{i=1}^N e_i \quad (2)$$

2.3.1 THE STATISTICAL PRINCIPLE OF TEAM SUPERIORITY

To show why an averaged team judgment is expected to be more accurate than a typical individual (Gigone & Hastie, 1997; Armstrong, 2001), we analyse the expected value and the variance of the collective error $\bar{e}_N$. In modelling human team performance, the theory relies on i) unbiased estimates where on average, individual human errors balance out to zero across the crowd: $\mathbb{E}[e_i] = 0$ for all $i \in \{1, \dots, N\}$ and ii) independence where team members do not influence one another, meaning individual errors are statistically uncorrelated: $\text{Cov}(e_i, e_j) = 0$ for all $i \neq j$. If each individual's judgment error has a variance of $\text{Var}(e_i) = \sigma^2$, the variance of the team's collective estimate is calculated as: $\text{Var}(\bar{e}_N) = \frac{1}{N^2} \sum_{i=1}^N \text{Var}(e_i) = \sigma^2/N$, so that the standard error of the team's judgment shrinks at the rate $\text{SE}(\bar{e}_N) = \sigma/\sqrt{N}$. As team size N grows, the variance approaches zero ($\lim_{N \rightarrow \infty} \text{Var}(\bar{e}_N) = 0$). This shows that the averaged team judgment will systematically be more accurate and consistent than any single individual judgment (Clemen, 1989; Page, 2007).

2.4 EFFECTIVE TEAM CAPACITY AND SCALING LIMITS ACROSS TASK TYPES

Comparing the mathematical foundations of the ideal disjunctive model (Section 2.2) and the ideal compensatory model (Section 2.3) reveals the theoretical origin of the scaling dichotomy: the failure probability of independent agents decreases exponentially in disjunctive tasks while it drops polynomially in compensatory tasks. More details of what behavior is expected from correlated and biased agents on disjunctive and compensatory task types are available in Appendix B.1. For correlated draws (i.e. answers sampled from a language model), the Kish design effect (Kish, 1965) provides the effective team capacity N_{eff} defined as the size of an ideal, independent ensemble ($\rho = 0$) that achieves the equivalent variance reduction as a correlated ensemble of size N : $N_{\text{eff}}(N) = N/(1 + (N - 1)\rho)$. For any positive pairwise error correlation $\rho > 0$: $\lim_{n \rightarrow \infty} N_{\text{eff}}(N) = 1/\rho$.

An important consequence arises when agents are independent samples from the same model, as in our experiments. Conditional on the item k , the outputs of such agents are independent and identically distributed with an item-specific distribution P_k . The pairwise correlation of their correctness across items then equals $\rho = \text{Var}_k(p_k)/(\bar{p}(1 - \bar{p}))$, where p_k is the single-agent success probability on item k and $\bar{p}$ its mean (Boland, 1989; Ladha, 1992); that is, ρ measures heterogeneity of item difficulty rather than interaction between agents. Such a team can be seen as a "crowd within" a single model, whose repeated judgments are more correlated than those of different individuals (Vul & Pashler, 2008; Herzog & Hertwig, 2009), and correlated judges add little information (Hogarth, 1978).

Proposition 2.1 (Task-type scaling under correlation and bias). *Let agents be conditionally independent given the item, and let $N_{\text{eff}}(N) \leq 1/\rho$ be defined as above.*

Disjunctive scaling regime. *With an oracle verifier, accuracy is $1 - \mathbb{E}_k[(1 - p_k)^N]$, which increases to the fraction of items with $p_k > 0$. Under plurality voting, the failure probability on item k decays exponentially in N if the correct answer is the unique mode of P_k and tends to one if it is not, so that accuracy converges to the fraction π of items on which the correct answer is the modal answer, that is, of the easy items of Chen et al. (2024). For binary outcomes, the attainable voting gain satisfies*

$$|\pi - \bar{p}| \leq 2\bar{p}(1 - \bar{p})(1 - \rho). \quad (3)$$

The gain from voting is therefore small when ρ is large, even though the exponential convergence of Eq. 1 holds on every item with $p_k > \frac{1}{2}$.

Compensatory scaling regime. *For continuous estimates with systematic bias b , variance σ^2 , and equicorrelated errors with correlation ρ , the mean squared error of the geometric mean is linear in $1/N_{\text{eff}}$:*

$$\text{MSE}_{\text{comp}}(N) = b^2 + \frac{\sigma^2}{N_{\text{eff}}(N)} \xrightarrow{N \rightarrow \infty} b^2 + \sigma^2 \rho. \quad (4)$$

With item-level biases b_k , the same decomposition reads $\text{MSE}(N) = \mathbb{E}_k[b_k^2] + \mathbb{E}_k[\sigma_k^2]/N$, so the attainable reduction is the within-item share of error, $1 - \beta$ with $\beta = \mathbb{E}[b_k^2]/(\mathbb{E}[b_k^2] + \mathbb{E}[\sigma_k^2])$. For example, with $\rho \approx 0.9$ (effective capacity $N_{\text{eff}} \approx 1.1$), averaging can remove at most $\sigma^2(1 - \rho) \approx 0.1 \sigma^2$, which is small relative to a dominant bias term b^2 . Proofs in Appendix B.3.

3 EVALUATION OF DISJUNCTIVE VS. COMPENSATORY SCALING

3.1 EXPERIMENTAL SETUP

To evaluate empirically whether LLM multi-agent team performance follows Proposition 2.1, we generated a corpus of approximately 6.8×10^7 per-agent generations for homogeneous and heterogeneous teams, evaluating all models on the same grid of team sizes $N \in \{1, 2, 3, 5, 7, 10, 15, 20, 25, 30\}$ and communication rounds (Rounds 1 to 3).

Models: We evaluate 13 open-weight models between 3B and 20B parameters, ordered by parameter count: smollm3-3b, qwen2.5-3b, qwen3-4b, mistral-7b, olmo2-7b, qwen2.5-7b, llama3.1-8b, marin-8b, qwen3-8b, phi4-14b, qwen3-14b, r1-distill-qwen-14b, and gpt-oss-20b. Unless stated otherwise, averages are macro-averages over the 13 models.

Benchmark Tasks: *Disjunctive tasks:* GSM8K (Cobbe et al., 2021) (grade-school mathematics), GSM-Hard (Gao et al., 2023) (GSM8K problems with larger numbers), MATH-500 (Hendrycks et al., 2021b) (competition mathematics), MMLU-Hard, a difficult subset of MMLU (Hendrycks et al., 2021a) (multitask knowledge), and ARC-Challenge (Clark et al., 2018) (science question answering). *Compensatory task:* RealFP Fermi problems (Kalyan et al., 2021), which require numerical order-of-magnitude estimation across physics, geography, and demographics.

Inference, Deliberation, and Metric Computation Protocol. All models are decoded at temperature $T = 0.4$, with three runs per configuration on the full test sets (Appendix C). The prompts request the final answer *before* a brief rationale (Appendix C); this answer-first format isolates the answer distribution of the instruction-tuned model itself rather than that of a test-time reasoning procedure, and makes answers robust to truncation. Two models, gpt-oss-20b and r1-distill-qwen-14b, reason before the formatted answer regardless of the prompt and thus provide a contrast between the two regimes. In Round 1, agents answer independently; in Rounds 2 and 3, each agent sees its own and its teammates’ previous responses and may revise its answer. Disjunctive answers are aggregated by plurality voting, and we additionally report pass@ N ; Fermi estimates are aggregated by the geometric mean (Eq. 2) and scored by the absolute log-error (ALE) and by within-one-order-of-magnitude accuracy. We clipped ALE at ± 3 orders of magnitude so that implausible reference labels (Section 3.2) and occasional runaway estimates do not dominate the mean. The conclusions

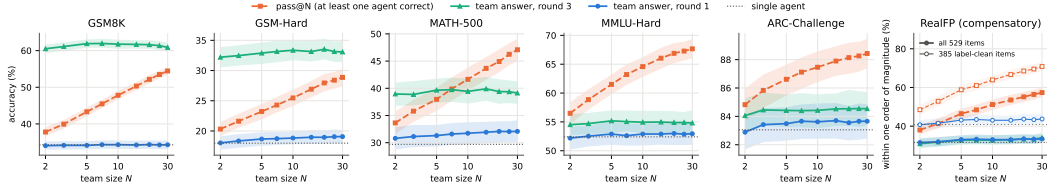


Figure 1: Scaling with team size (average over 13 models; bands are 95% item-bootstrap intervals). The team answer is the plurality vote on the disjunctive tasks and the geometric mean on a representative compensatory RealFP (right), where accuracy means an estimate within one order of magnitude. Exact values are given in Appendix Table 4.

do not depend on this threshold (Appendix Table 7). For RealFP, we analyse the runs generated with the decomposition prompt (10 of the 13 models; Appendix C). Confidence intervals are 95% item-bootstrap intervals.

3.2 EMPIRICAL SCALING DYNAMICS ON DISJUNCTIVE VS COMPENSATORY TASKS

Disjunctive From Fig. 1 and Table 4 we can see that the probability that at least one of N independent agents is correct (pass@ N , Section 2.2) rises from the single-agent accuracy to pass@30 of 54.4% on GSM8K (single agent: 34.3%), 47.1% on MATH-500 (29.7%), 67.7% on MMLU-Hard (52.4%), 28.9% on GSM-Hard (18.0%), and 88.4% on ARC-Challenge (83.1%). In Steiner’s terms, the potential productivity of the team increases with its size. Allowing agents to revise their answers over multiple rounds leads to large accuracy gains on arithmetic tasks. GSM8K accuracy nearly doubles from 34.1% (Round 1, $N = 5$) to 61.9% (Round 3, $N = 5$); GSM-Hard accuracy rises from 18.7% to 32.9% and MATH-500 accuracy from 31.3% to 39.7%. The gain is, however, nearly independent of team size: on GSM8K it is 26.5 points with a single peer ($N = 2$) and 26.6 points with 29 peers (paired difference -0.1 , 95% CI $[-0.6, 0.3]$). Because the Round-1 format asks for the answer before the rationale, a substantial part of this gain is likely due to the opportunity to reason before committing to an answer; a single agent given a five-fold token budget improves by 12.4 points on GSM8K (Appendix C.1 shows an example).

On **compensatory** problems, of which Fermi are representative, Fig. 1 and Table 4 show that increasing team size N provides little improvement in either round. Under Round 1 averaging, accuracy within one order of magnitude rises from 31.6% for a single agent to 33.1% at $N = 7$ and 33.4% at $N = 30$, and the clipped log-error decreases from 1.84 to 1.73 at $N = 5$ and 1.71 at $N = 30$. Revision does not help: Round-3 accuracy differs from Round-1 accuracy by between -0.6 and $+0.7$ points across team sizes. The potential of Fermi teams nevertheless grows: the proportion of items on which at least one agent is within one order of magnitude rises from 31.6% for a single agent to 57.3% at $N = 30$ (Section 2.3; Appendix Table 4), but averaging cannot select that estimate. Continuous estimates lack verifiable intermediate steps, so revision rarely corrects erroneous orders of magnitude. RealFP contains implausible reference values, such as 2×10^{90} for the number of a person’s ancestors; on 144 of the 529 items (27.2%), the median estimate of the ten models deviates from the reference by more than three orders of magnitude. On the remaining 385 items (open markers in Fig. 1; fermi-clean in Table 4), all levels are higher, but the pattern is unchanged: Round-1 accuracy rises only from 40.9% for a single agent to 43.6% at $N = 30$ and the clipped log-error decreases from 1.52 to 1.39 at $N = 5$ and 1.37 at $N = 30$, whereas pass@30 reaches 70.8%. The gap between the potential and the realised performance of Fermi teams therefore does not arise from erroneous labels.

3.3 THE ANATOMY OF THE DISJUNCTIVE CEILING

While Section 3.2 showed that disjunctive tasks benefit substantially from multi-agent deliberation compared to compensatory tasks, a critical question remains: *does disjunctive performance scale monotonically as team size grows?* Figure 1 shows that it does not: rather

than approaching the pass@ N curve, realised disjunctive accuracy exhibits its own ceiling. In this section, we analyse the empirical evidence and the mechanisms governing this ceiling.

The Early Saturation of Independent Ensembles (Round 1) Round-1 plurality voting yields almost no improvement with team size: from $N = 2$ to $N = 30$, accuracy changes by +0.3 points on GSM8K (95% CI [0.0, 0.5]), +1.0 on GSM-Hard, +1.3 on MATH-500, +0.8 on MMLU-Hard, and +0.8 on ARC-Challenge. As Proposition 2.1 predicts, plurality voting converges to the model’s modal answer on each item. We estimated the item-level answer distributions from the Round-1 answers of the $N = 30$ teams (up to 90 samples per item) and predicted plurality accuracy at every team size by resampling. Across 650 model–task–team-size configurations, predicted and observed accuracies differ by 0.48 points on average (median 0.18; Pearson $r = 0.999$; Appendix Figure 2). The intra-class correlation of correctness ranges from 0.53 to 0.97 (mean 0.78), and the predicted large-team limit exceeds single-agent accuracy by only 0.4–1.2 points per task. Consistent with Eq. 3, the gain remains within the bound $2\bar{p}(1 - \bar{p})(1 - \rho)$ for all 65 model–task pairs and uses on average only 11–21% of it (Appendix Figure 3). The two models that reason before answering have a lower ρ (0.66 against 0.81), a larger predicted voting gain (+3.4 against +0.2 points), and observed gains of +7.2 and +2.8 points from one to 30 agents, against -1.0 to $+1.0$ for the others, consistent with Eq. 3 and with the success of self-consistency under chain-of-thought prompting (Wang et al., 2023); the model predicts both regimes (mean error 0.19 and 0.43 points, excluding MATH-500). A higher temperature is no substitute: for qwen2.5-7b, raising T from 0.2 to 1.0 lowers ρ and raises pass@30 but leaves plurality accuracy unchanged, because the extra samples spread over incorrect answers (Appendix Table 5). Additional queries cannot produce a correct plurality on items whose modal answer is wrong; the difference between pass@30 and plurality accuracy at $N = 30$ (4.8–20.1 points) is the process loss of plurality voting.

The Peak-and-Decay Regime in Deliberated Teams (Round 3) Multi-turn communication nearly doubles accuracy on gsm8k (34.1% $\rightarrow$ 62.0% at the peak) and gsmhard (18.0% $\rightarrow$ 33.5%). Deliberated teams do not, however, continue to improve with size (Appendix Table 6): accuracy peaks at an intermediate team size and declines by 0.4–1.1 points towards $N = 30$ on the mathematical and knowledge tasks, most clearly on gsm8k (-1.0 points relative to $N = 5$, 95% CI $[-1.4, -0.7]$), and is flat on arc. The largest declines occur on multi-step symbolic tasks, consistent with an incorrect intermediate result shared by several peers acting as an attractor during revision.

Per-Model Scaling and Correlation Breakdown Deliberation gains at $N = 5$ range from +2.8 (mistral-7b) to +20.9 points (phi4-14b; Table 2) and are not explained by ρ (rank correlation 0.12, $p = 0.71$). The two strongest models, gpt-oss-20b and r1-distill-qwen-14b, gain little (+3.2 and +6.7 points), since they reason before the formatted answer.

Asymptotic Capacity Saturation ($N_{\text{eff}} \leq 1/\rho$) and Post-Peak Decline ($\Delta_{\text{decay}} < 0$): With ρ between 0.64 and 0.93, $N_{\text{eff}}(5)$ lies between 1.06 and 1.40 and $N_{\text{eff}}(30)$ between 1.07 and 1.53, so a six-fold larger team adds almost no effective capacity. Scaling from $N = 5$ to $N = 30$ significantly reduces Round-3 accuracy in 6 of the 13 models (by up to -3.01 points for marin-8b, 95% CI $[-3.71, -2.33]$, and up to -6.1 points on GSM8K), significantly increases it in three, and leaves it unchanged in four.

On Architectural Diversity The last rows of Table 2 evaluate majority voting over the single-agent answers of five different models, choosing the best of the $\binom{8}{5} = 56$ combinations of the eight strongest panel models on a random half of the items and evaluating it on the other half. The most frequently selected pool (gpt-oss-20b, r1-distill-qwen-14b, qwen3-8b, phi4-14b, smollm3-3b) has a pairwise error correlation of $\rho = 0.37$ ($N_{\text{eff}}(5) = 2.01$) and reaches 83.2% on held-out items, +22.1 points above the mean of its members, but it falls short of its strongest member (gpt-oss-20b, 86.8%) in all 10 splits, by 3.4 to 4.3 points, consistent with Pappu et al. (2026); equal-weight voting is suboptimal when members differ in competence (Nitzan & Paroush, 1982). Furthermore, five 7B–8B models from different developers (qwen2.5-7b, llama3.1-8b, mistral-7b, olmo2-7b, marin-8b) have $\rho = 0.54$ ($N_{\text{eff}} = 1.58$) and reach 39.8%, +7.8 points above the mean member (95% CI [7.0, 8.5]) but 2.5 points below the strongest member, qwen2.5-7b ([1.2, 3.8]). A real heterogeneous run

Table 2: **Per-Model Scaling and Correlation Breakdown on Disjunctive Tasks** (Round 3; accuracy in %, macro-averaged over the five benchmarks). $\Delta_5 = \text{Acc}_5 - \text{Solo}$ and $\Delta_{\text{decay}} = \text{Acc}_{30} - \text{Acc}_5$, with paired 95% item-bootstrap intervals in brackets; ρ is the intra-class correlation of correctness. [†]Best five-model pool, selected and evaluated on disjoint halves of the items (mean over 10 splits; brackets give the range across splits); for pools, ρ is the mean correlation between members.

Model	Solo Acc	N = 5 Acc	N = 30 Acc	Δ_5 Gain	Δ_{decay}	ρ	N_{eff} (5)	N_{eff} (30)
Homogeneous Architectures (Multi-Turn Deliberation, Round 3)								
smollm3-3b	32.43	45.09	45.61	+12.66 [+11.80, +13.55]	+0.52 [-0.11, +1.15]	0.67	1.36	1.47
qwen2.5-3b	27.08	32.32	32.58	+5.23 [+4.57, +5.88]	+0.26 [-0.31, +0.80]	0.82	1.16	1.20
qwen3-4b	42.13	51.21	52.71	+9.09 [+8.34, +9.87]	+1.50 [+1.05, +1.97]	0.86	1.13	1.16
mistral-7b	26.14	28.96	27.34	+2.82 [+2.34, +3.30]	-1.62 [-2.09, -1.19]	0.84	1.15	1.18
olmo2-7b	27.17	41.73	39.26	+14.56 [+13.54, +15.46]	-2.47 [-3.16, -1.77]	0.74	1.26	1.34
qwen2.5-7b	40.49	53.61	52.29	+13.12 [+12.12, +14.06]	-1.32 [-1.88, -0.78]	0.91	1.08	1.10
llama3.1-8b	32.47	50.22	49.44	+17.75 [+16.80, +18.71]	-0.78 [-1.53, -0.01]	0.77	1.23	1.29
marin-8b	28.62	45.75	42.74	+17.14 [+16.19, +18.03]	-3.01 [-3.71, -2.33]	0.64	1.40	1.53
qwen3-8b	51.02	61.90	63.21	+10.89 [+10.06, +11.67]	+1.31 [+0.77, +1.85]	0.81	1.18	1.22
phi4-14b	48.40	69.30	68.20	+20.90 [+19.86, +21.99]	-1.10 [-1.66, -0.49]	0.87	1.12	1.15
qwen3-14b	50.19	63.37	63.71	+13.18 [+12.28, +14.05]	+0.34 [-0.08, +0.79]	0.93	1.06	1.07
r1-distill-	74.50	81.17	83.01	+6.68 [+5.64, +7.64]	+1.84 [+1.13, +2.53]	0.64	1.40	1.53
gpt-oss-20b	84.72	87.87	88.24	+3.16 [+2.62, +3.71]	+0.37 [-0.00, +0.75]	0.68	1.34	1.45
Heterogeneous Combinations (Single-Agent Answers, Majority Voting)								
Hetero Best [†]	61.02	83.16	—	+22.14 [+21.36, +22.74] [†]	—	0.37	2.01	—
Hetero 7B-8B	31.93	39.76	—	+7.83 [+7.02, +8.52]	—	0.54	1.58	—

with the same models (100 items per task) confirms this (22.2% against 30.3% for qwen2.5-7b at $N = 5$).

3.4 THE ANATOMY OF THE COMPENSATORY CEILING

Section 3.2 showed that compensatory tasks benefit little from multi-agent scaling or deliberation. Table 3 provides a per-model breakdown. For homogeneous models, the error reduction from a single agent to a team of five ranges from 0.9% to 8.1%, with one exception (marin-8b, 15.4%), and increasing the team to $N = 30$ yields little further reduction. The intra-class correlation of signed log-errors lies between 0.70 and 0.95, and effective team capacity $N_{\text{eff}}(5)$ lies between 1.04 and 1.32: five agents provide the variance reduction of little more than one. Averaged over models, item-level bias accounts for $\beta = 0.87$ of the mean squared log-error (between-item standard deviation of b_k : 1.96 orders of magnitude; within-item standard deviation: 0.75), so that even an infinite homogeneous team could remove at most about 13% of it. Eq. 4 predicts the observed error of the geometric mean at every team size to within 0.6% on average for all ten models (Pearson $r = 0.997$; Appendix Figure 4). Only the model with the most dispersed samples (marin-8b) and heterogeneous multi-family teams depart noticeably from the ceiling: the 7B-8B pool reduces the error of its average member by 25.8% (95% CI [23.1, 28.7]) and, with 1.45, is also more accurate than its best member alone (1.80). **Temperature Invariance** For four models at $T \in \{0.4, 0.8, 1.0\}$ (Appendix Table 8), Fermi accuracy is virtually identical across temperatures: higher temperature increases the within-item dispersion of log-errors by 30-70%, but the intra-class correlation remains between 0.83 and 0.98.

Empirical Analysis of Unbiasedness The classical model in Section 2.3 assumes that errors balance out to zero on average ($\mathbb{E}[e_i] = 0$). To test this condition, we measure the distribution of signed log-errors $e_i = \log_{10}(x_i/g)$ in Appendix Table 9. Models such as mistral-7b ($b = +0.71$), qwen2.5-3b ($b = +0.65$), and llama3.1-8b ($b = +0.62$) overestimate by more than half an order of magnitude on average, and 61-64% of their estimates exceed the ground truth. Conversely, marin-8b ($b = -0.20$, 95% CI [-0.38, -0.03]) significantly underestimates. Even for the models whose global mean is not significantly different from zero (qwen2.5-7b, $b = +0.11$, [-0.05, 0.28], and qwen3-8b, $b = -0.04$, [-0.20, 0.13]), the unbiasedness condition

Table 3: **Per-Model Scaling Breakdown on Compensatory Fermi Estimation** (RealFP, 529 items, Round 1, decomposition-prompt runs). MAE is the mean absolute log-error of the geometric mean (clipped at ± 3); “Drop” is its relative reduction from $N = 1$ to $N = 5$, with a paired 95% item-bootstrap interval; ρ is the intra-class correlation of signed log-errors. [‡]The five 7B–8B models of Section 3.3, one answer each ($N = 5$) or six each ($N = 30$); “Solo” is their mean, and ρ their mean cross-model bias correlation.

Model	Solo MAE	N = 5 MAE	N = 30 MAE	Drop %	Solo Acc	N = 5 Acc	ρ	N _{eff} (5)
smollm3-3b	1.85	1.71	1.67	+7.3 [+4.7, +9.8]	31.9	33.1 [29.4, 36.8]	0.81	1.18
qwen2.5-3b	1.94	1.86	1.84	+4.1 [+2.0, +6.1]	28.6	26.1 [22.9, 29.5]	0.87	1.12
mistral-7b	2.08	1.91	1.91	+8.1 [+6.1, +11.0]	23.4	27.7 [23.6, 31.6]	0.87	1.12
olmo2-7b	1.80	1.70	1.68	+5.5 [+3.1, +7.3]	34.1	34.5 [31.3, 38.7]	0.87	1.11
qwen2.5-7b	1.80	1.72	1.71	+3.9 [+2.3, +5.6]	34.0	33.9 [30.1, 37.7]	0.90	1.08
llama3.1-8b	2.02	1.87	1.84	+7.9 [+5.0, +10.7]	23.7	26.7 [23.1, 30.2]	0.84	1.15
marin-8b	2.07	1.75	1.69	+15.4 [+13.0, +17.7]	24.1	31.2 [27.9, 34.6]	0.70	1.32
qwen3-8b	1.64	1.61	1.60	+2.2 [+0.2, +4.8]	38.7	38.4 [33.8, 42.5]	0.95	1.04
phi4-14b	1.66	1.61	1.61	+2.7 [+0.6, +5.0]	39.2	41.0 [37.1, 45.6]	0.94	1.05
qwen3-14b	1.59	1.57	1.57	+0.9 [-1.6, +3.1]	38.4	38.6 [34.4, 42.7]	0.93	1.06
Heterogeneous:								
Hetero 7B–8B [‡]	1.95	1.45	1.41	+25.8 [+23.1, +28.7]	27.8	38.4 [34.2, 42.3]	0.61	1.45

fails at the level of individual items. For a given prompt k , independent samples from the same model cluster around the model’s item-specific estimate, so that $\mathbb{E}[e_i \mid \text{item } k] = b_k \neq 0$. The root-mean-square item-level bias, averaged across models, is 2.01 orders of magnitude (with clipping at ± 3). Intra-model averaging reduces sampling variation around b_k but leaves the systematic error b_k untouched (Appendix C.1 shows an example).

Cross-Model Error Correlation and Shared Directional Misconceptions. If individual models have non-zero item-level biases, can a multi-model ensemble cancel these errors? That is, does $\sum_{m=1}^M \mathbb{E}[e^{(m)} \mid \text{item } k] \approx 0$? To answer this question, we compute the pairwise correlation of item-level biases across models (Appendix Table 10). The mean pairwise correlation is $r = 0.63$ (95% CI [0.60, 0.67]; standard deviation across model pairs 0.06), ranging from $r = 0.46$ between mistral-7b and qwen3-8b to $r = 0.77$ between qwen3-8b and qwen3-14b. On 31.5% of all items, *every model errs in the same direction*. Because these correlations are well below one, averaging across model families can cancel part of the item-level bias, which is consistent with the gains of the heterogeneous pool in Table 3. Part of this alignment is due to erroneous reference labels: on the 385 label-clean items (Section 3.2), the mean correlation falls to $r = 0.41$ and all models err in the same direction on 19.1% of items, whereas the bias share remains high ($\beta = 0.82$).

4 DISCUSSION

Implications. On disjunctive tasks (Section 3.3), the difference between pass@ N and plurality accuracy identifies the process loss that verifiers, reward models, or structured deliberation must recover; reasoning before answering or combining model families can raise the voting ceiling, by an amount that Eq. 3 bounds. On compensatory tasks (Section 3.4), reducing the item-level bias requires changing the estimates themselves, for instance by combining models whose biases are not aligned, rather than adding samples; prompting interventions and a fine-tuned model are evaluated in Appendix F.

Limitations. First, the prompts request the answer before the rationale, which lowers absolute accuracy and shapes the deliberation gain; chain-of-thought prompting was not evaluated for the answer-first models, so the size of the voting ceiling is specific to this regime, whereas its mechanism is not. Second, there is no revision condition without peers. Third, the Fermi results rest on one benchmark with substantial label noise, generated with two prompt families (Appendix C). Fourth, all models are open-weight with at most 20B parameters, and additive, conjunctive, and discretionary tasks remain to be studied.

REPRODUCIBILITY STATEMENT

The theoretical results rest on the assumptions stated in Proposition 2.1; the ideal-case analysis and complete proofs are given in Appendix B.1 and Appendix B.3. Models (with Hugging Face identifiers), prompts, decoding parameters, the number of runs per configuration, and the benchmark items used are specified in Appendix C, and the data-processing steps, including the identification of the RealFP prompt families and the corrections and exclusions of the log audit, in the data-integrity paragraph of the same appendix. The metrics, the clipping of log-errors, the label-clean RealFP subset, and the item-bootstrap intervals are defined in Section 3, with sensitivity analyses in Appendices D and E. The per-agent generation logs will be released upon publication; the analysis scripts, which reproduce all tables and figures from these logs, will be open sourced upon acceptance.

USE OF LARGE LANGUAGE MODELS

The research question, the application of Steiner’s taxonomy to multi-agent LLM systems, and the design and execution of the experiments are the authors’ own. We used generative AI tools (a coding and writing assistant) to provide feedback on the methodology in the form of a simulated review; to help correct and refine the theoretical analysis, including the voting-gain bound (Eq. 3), the compensatory error decomposition (Eq. 4), and parts of their proofs; to implement analysis code, including the bootstrap intervals, the validation of Eqs. 3 and 4, the identification of prompt families and implausible reference labels, and the sensitivity analyses; and to draft interpretations of the results. We did not use generative AI tools to generate synthetic data, and qualitative or thematic data analysis is not applicable to this work. Additionally, we used these tools to create figures, draft and edit parts of the text, suggest relevant literature, and format and check references. All references, all sections of the paper including text, were revised manually by all authors. AI-generated code was reviewed by the authors, and all reported numbers were recomputed from the raw generation logs; every suggested reference was checked against its published source. We take responsibility for the final content of this work, including text, claims, and artifacts produced with the aid of generative AI.

REFERENCES

- J. Scott Armstrong (ed.). *Principles of Forecasting: A Handbook for Researchers and Practitioners*, volume 30 of *International Series in Operations Research & Management Science*. Kluwer Academic Publishers, Boston, MA, 2001. doi: 10.1007/978-0-306-47630-3.
- Blaž Bertalanč and Carolina Fortuna. The Ringelmann effect in multi-agent LLM systems: A scaling law for effective team size. *arXiv preprint arXiv:2606.02646*, 2026.
- Philip J. Boland. Majority systems and the Condorcet jury theorem. *Journal of the Royal Statistical Society: Series D (The Statistician)*, 38(3):181–189, 1989. doi: 10.2307/2348873.
- Chi-Min Chan, Weize Chen, Yusheng Su, Jianxuan Yu, Wei Xue, Shanghang Zhang, Jie Fu, and Zhiyuan Liu. ChatEval: Towards better LLM-based evaluators through multi-agent debate. In *International Conference on Learning Representations (ICLR)*, 2024.
- Lingjiao Chen, Jared Davis, Boris Hanin, Peter Bailis, Ion Stoica, Matei Zaharia, and James Zou. Are more lm calls all you need? towards the scaling properties of compound ai systems. In *Proceedings of the 38th International Conference on Neural Information Processing Systems*, NIPS ’24, Red Hook, NY, USA, 2024. Curran Associates Inc. ISBN 9798331314385.
- Yun-Shiuan Chuang, Sameer Narendran, Nikunj Harlalka, Alexander Cheung, Sizhe Gao, Siddharth Suresh, Junjie Hu, and Timothy T. Rogers. Probing LLM world models: Enhancing guesstimation with wisdom of crowds decoding. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 4699–4713. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.234.

- Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? Try ARC, the AI2 reasoning challenge. *arXiv preprint arXiv:1803.05457*, 2018.
- Robert T. Clemen. Combining forecasts: A review and annotated bibliography. *International Journal of Forecasting*, 5(4):559–583, 1989. doi: 10.1016/0169-2070(89)90012-5.
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training verifiers to solve math word problems. *arXiv preprint arXiv:2110.14168*, 2021.
- Clinton P. Davis-Stober, David V. Budescu, Jason Dana, and Stephen B. Broomell. When is a crowd wise? *Decision*, 1(2):79–101, 2014. doi: 10.1037/dec0000004.
- Yilun Du, Shuang Li, Antonio Torralba, Joshua B. Tenenbaum, and Igor Mordatch. Improving factuality and reasoning in language models through multiagent debate. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, volume 235 of *Proceedings of Machine Learning Research*, pp. 11733–11763, 2024.
- Elliot Epstein, John Winnicki, Thanawat Sornwanee, and Rajat Vadraj Dwaraknath. LLMs are overconfident: Evaluating confidence interval calibration with FermiEval. *arXiv preprint arXiv:2510.26995*, 2025.
- Francis Galton. Vox populi. *Nature*, 75(1949):450–451, 1907. doi: 10.1038/075450a0.
- Luyu Gao, Aman Madaan, Shuyan Zhou, Uri Alon, Pengfei Liu, Yiming Yang, Jamie Callan, and Graham Neubig. PAL: Program-aided language models. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, volume 202 of *Proceedings of Machine Learning Research*, pp. 10764–10799, 2023.
- Daniel Gigone and Reid Hastie. Proper analysis of the accuracy of group judgments. *Psychological Bulletin*, 121(1):149–167, 1997. doi: 10.1037/0033-2909.121.1.149.
- Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding. In *International Conference on Learning Representations (ICLR)*, 2021a.
- Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, and Jacob Steinhardt. Measuring mathematical problem solving with the MATH dataset. In *Proceedings of the Neural Information Processing Systems Track on Datasets and Benchmarks*, volume 1, 2021b.
- Stefan M. Herzog and Ralph Hertwig. The wisdom of many in one mind: Improving individual judgments with dialectical bootstrapping. *Psychological Science*, 20(2):231–237, 2009. doi: 10.1111/j.1467-9280.2009.02271.x.
- Robin M. Hogarth. A note on aggregating opinions. *Organizational Behavior and Human Performance*, 21(1):40–46, 1978. doi: 10.1016/0030-5073(78)90037-5.
- Alan G. Ingham, George Levinger, James Graves, and Vaughn Peckham. The Ringelmann effect: Studies of group size and group performance. *Journal of Experimental Social Psychology*, 10(4):371–384, 1974. doi: 10.1016/0022-1031(74)90033-X.
- Carlos E Jimenez, John Yang, Alexander Wettig, Shunyu Yao, Kexin Pei, Ofir Press, and Karthik R Narasimhan. SWE-bench: Can language models resolve real-world github issues? In *The Twelfth International Conference on Learning Representations*, 2024. URL <https://openreview.net/forum?id=VTF8yNQm66>.
- junyou li, Qin Zhang, Yangbin Yu, QIANG FU, and Deheng Ye. More agents is all you need. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=bgzUSZ8aeg>.

- Ashwin Kalyan, Abhinav Kumar, Arjun Chandrasekaran, Ashish Sabharwal, and Peter Clark. How much coffee was consumed during EMNLP 2019? Fermi problems: A new reasoning challenge for AI. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 7318–7328. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.emnlp-main.582.
- Ezra Karger, Houtan Bastani, Chen Yueh-Han, Zachary Jacobs, Danny Halawi, Fred Zhang, and Philip Tetlock. Forecastbench: A dynamic benchmark of AI forecasting capabilities. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=1fPkGWXLlf>.
- Norbert L. Kerr and Steven E. Bruun. Dispensability of member effort and group motivation losses: Free-rider effects. *Journal of Personality and Social Psychology*, 44(1):78–94, 1983. doi: 10.1037/0022-3514.44.1.78.
- Elliot Myunghoon Kim, Avi Garg, Kenny Peng, and Nikhil Garg. Correlated errors in large language models. In *Proceedings of the 42nd International Conference on Machine Learning (ICML)*, volume 267 of *Proceedings of Machine Learning Research*, pp. 30038–30066, 2025.
- Yubin Kim, Ken Gu, Chanwoo Park, Chunjong Park, Samuel Schmidgall, A. Ali Heydari, Yao Yan, Zhihan Zhang, Yuchen Zhuang, Yun Liu, Mark Malhotra, Paul Pu Liang, Hae Won Park, Yuzhe Yang, Xuhai Xu, Yilun Du, Shwetak Patel, Tim Althoff, Daniel McDuff, and Xin Liu. Capable language models can outgrow the benefits of collaboration. *Nature Machine Intelligence*, 8(7):1157–1172, 2026. ISSN 2522-5839. doi: 10.1038/s42256-026-01268-y. URL <https://doi.org/10.1038/s42256-026-01268-y>.
- Leslie Kish. *Survey Sampling*. John Wiley & Sons, New York, 1965.
- Krishna K. Ladha. The Condorcet jury theorem, free speech, and correlated votes. *American Journal of Political Science*, 36(3):617–634, 1992. doi: 10.2307/2111584.
- Richard P. Larrick and Jack B. Soll. Intuitions about combining opinions: Misappreciation of the averaging principle. *Management Science*, 52(1):111–127, 2006. doi: 10.1287/mnsc.1050.0459.
- Bibb Latané, Kipling Williams, and Stephen Harkins. Many hands make light the work: The causes and consequences of social loafing. *Journal of Personality and Social Psychology*, 37(6):822–832, 1979. doi: 10.1037/0022-3514.37.6.822.
- Patrick R. Laughlin and Alan L. Ellis. Demonstrability and social combination processes on mathematical intellectual tasks. *Journal of Experimental Social Psychology*, 22(3):177–189, 1986. doi: 10.1016/0022-1031(86)90022-3.
- Guohao Li, Hasan Abed Al Kader Hammoud, Hani Itani, Dmitrii Khizbullin, and Bernard Ghanem. Camel: communicative agents for "mind" exploration of large language model society. In *Proceedings of the 37th International Conference on Neural Information Processing Systems, NIPS '23*, Red Hook, NY, USA, 2023. Curran Associates Inc.
- Xiao Liu, Hao Yu, Hanchen Zhang, Yifan Xu, Xuanyu Lei, Hanyu Lai, Yu Gu, Hangliang Ding, Kaiwen Men, Kejuan Yang, Shudan Zhang, Xiang Deng, Aohan Zeng, Zhengxiao Du, Chenhui Zhang, Sheng Shen, Tianjun Zhang, Yu Su, Huan Sun, Minlie Huang, Yuxiao Dong, and Jie Tang. AgentBench: Evaluating LLMs as agents. In *International Conference on Learning Representations (ICLR)*, 2024.
- Zishuo Liu, Carlos Rabat Villarreal, Mostafa Rahgouy, Amit Das, Zheng Zhang, Chang Ren, and Dongji Feng. LLM for complex reasoning task: An exploratory study in Fermi problems. *arXiv preprint arXiv:2504.02671*, 2025.
- Shmuel Nitzan and Jacob Paroush. Optimal decision rules in uncertain dichotomous choice situations. *International Economic Review*, 23(2):289, 1982. doi: 10.2307/2526438.
- Jay A. Olson, Johnny Nahas, Denis Chmoulevitch, Simon J. Cropper, and Margaret E. Webb. Naming unrelated words predicts creativity. *Proceedings of the National Academy of Sciences*, 118(25):e2022340118, 2021. doi: 10.1073/pnas.2022340118.

- Scott E. Page. *The Difference: How the Power of Diversity Creates Better Groups, Firms, Schools, and Societies*. Princeton University Press, Princeton, NJ, 2007.
- Aneesh Pappu, Batu El, Hancheng Cao, Carmelo di Nolfo, Yanchao Sun, Meng Cao, and James Zou. Multi-agent teams hold experts back. In *Forty-third International Conference on Machine Learning*, 2026. URL <https://openreview.net/forum?id=Xn8kmKv09g>.
- David Rein, Betty Li Hou, Asa Cooper Stickland, Jackson Petty, Richard Yuanzhe Pang, Julien Dirani, Julian Michael, and Samuel R. Bowman. GPQA: A graduate-level google-proof q&a benchmark. In *First Conference on Language Modeling*, 2024. URL <https://openreview.net/forum?id=Ti67584b98>.
- Ivan D. Steiner. *Group Process and Productivity*. Academic Press, New York, 1972.
- James Surowiecki. *The Wisdom of Crowds: Why the Many Are Smarter Than the Few and How Collective Wisdom Shapes Business, Economies, Societies, and Nations*. Doubleday, New York, 2004.
- Edward Vul and Harold Pashler. Measuring the crowd within: Probabilistic representations within individuals. *Psychological Science*, 19(7):645–647, 2008. doi: 10.1111/j.1467-9280.2008.02136.x.
- Junlin Wang, Jue WANG, Ben Athiwaratkun, Ce Zhang, and James Zou. Mixture-of-agents enhances large language model capabilities. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=h0ZfDlrj7T>.
- Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc V Le, Ed H. Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2023. URL <https://openreview.net/forum?id=1PL1NIMMrw>.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, Tianle Li, Max Ku, Kai Wang, Alex Zhuang, Rongqi Fan, Xiang Yue, and Wenhui Chen. MMLU-pro: A more robust and challenging multi-task language understanding benchmark. In *The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2024. URL <https://openreview.net/forum?id=y10DM6R2r3>.
- Zhiyuan Weng, Guikun Chen, and Wenguan Wang. Do as we do, not as you think: the conformity of large language models. In *International Conference on Learning Representations (ICLR)*, 2025.
- Qingyun Wu, Gagan Bansal, Jieyu Zhang, Yiran Wu, Beibin Li, Erkang Zhu, Li Jiang, Xiaoyun Zhang, Shaokun Zhang, Jiale Liu, Ahmed Hassan Awadallah, Ryen W. White, Doug Burger, and Chi Wang. AutoGen: Enabling next-gen LLM applications via multi-agent conversations. In *Conference on Language Modeling (COLM)*, 2024. URL <https://openreview.net/forum?id=BAakY1hNKS>.
- Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, Siheng Zhao, Ruisheng Cao, Toh Jing Hua, Zhoujun Cheng, Dongchan Shin, Fangyu Lei, Yitao Liu, Yiheng Xu, Shuyan Zhou, Silvio Savarese, Caiming Xiong, Victor Zhong, and Tao Yu. Osworld: benchmarking multimodal agents for open-ended tasks in real computer environments. In *Proceedings of the 38th International Conference on Neural Information Processing Systems, NIPS ’24*, Red Hook, NY, USA, 2024. Curran Associates Inc. ISBN 9798331314385.
- Frank F. Xu, Yufan Song, Boxuan Li, Yuxuan Tang, Kritanjali Jain, Mengxue Bao, Zora Zhiruo Wang, Xuhui Zhou, Zhitong Guo, Murong Cao, Mingyang Yang, Hao Yang Lu, Amaad Martin, Zhe Su, Leander Melroy Maben, Raj Mehta, Wayne Chi, Lawrence Keunho Jang, Yiqing Xie, Shuyan Zhou, and Graham Neubig. Theagentcompany: Benchmarking LLM agents on consequential real world tasks. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems Datasets and Benchmarks Track*, 2026. URL <https://openreview.net/forum?id=LZnKNApvhG>.

Yingxuan Yang, Chengrui Qu, Muning Wen, Laixi Shi, Ying Wen, Weinan Zhang, Adam Wierman, and Shangding Gu. Understanding agent scaling in LLM-based multi-agent systems via diversity. *arXiv preprint arXiv:2602.03794*, 2026. URL <https://openreview.net/forum?id=9BN2W5BCfE>.

Terry Yue Zhuo, Vu Minh Chien, Jenny Chim, Han Hu, Wenhao Yu, Ratnadira Widyasari, Imam Nur Bani Yusuf, Haolan Zhan, Junda He, Indraneil Paul, Simon Brunner, Chen GONG, James Hoang, Armel Randy Zebaze, Xiaoheng Hong, Wen-Ding Li, Jean Kaddour, Ming Xu, Zhihan Zhang, Prateek Yadav, Naman Jain, Alex Gu, Zhoujun Cheng, Jiawei Liu, Qian Liu, Zijian Wang, David Lo, Binyuan Hui, Niklas Muennighoff, Daniel Fried, Xiaoning Du, Harm de Vries, and Leandro Von Werra. Bigcodebench: Benchmarking code generation with diverse function calls and complex instructions. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=YrycTj1lL0>.

A DETAILED STEINER TAXONOMY MAPPING ON ALL 5 TASK CATEGORIES

Disjunctive Tasks in which the outcome is established through $Y = \max_i \{y_i\}$ or plurality/majority/verifier selection and/or debate. The tasks are unitary and optimizing and group success requires that *at least one* agent discovers a valid solution, and the group then recognizes it as such. In recent literature, example benchmarks falling under this task type are: i) multi-step mathematical reasoning such as MATH-500 (Hendrycks et al., 2021b; Du et al., 2024) and GSM8K (Cobbe et al., 2021), ii) challenging multiple-choice knowledge suites such as MMLU-Pro (Wang et al., 2024) and GPQA Diamond (Rein et al., 2024), iii) school science answering in ARC-Challenge (Clark et al., 2018), and iv) functional code generation in BigCodeBench (Zhuo et al., 2025). Multi-agent LLM systems operationalize this task structure via plurality voting (Wang et al., 2023), verifier re-ranking, and multi-agent debate (Du et al., 2024; Wang et al., 2025), in some studies reporting success while in others showing diminishing benefits with scale.

Compensatory Tasks in which the outcome is determined by $Y = \text{Aggr}(x_1, \dots, x_n)$ via statistical aggregation (geometric mean, arithmetic mean, or median) of continuous numerical estimates ($x_i \in \mathbb{R}^+$) or calibrated probabilities ($p_i \in [0, 1]$). In recent AI research, this type is exemplified by post-cutoff future event forecasting in ForecastBench (Karger et al., 2025), numerical guesstimation of world-model properties in Guesstimate (Chuang et al., 2025), multi-scale physical and dimensional estimation in RealFP and SynthFP (Kalyan et al., 2021), and competitive Science Olympiad estimation in SciOly. LLMs typically perform worse on such tasks types compared to disjunctive types Liu et al. (2025); Epstein et al. (2025).

Additive Tasks in which the outcome is established through $Y = \sum_i y_i$ or $Y = \bigcup_i S_i$ via cumulative sum, total coverage, or deduplicated set union of independent member efforts. These tasks are divisible and maximizing by nature. This type represents the classical study by Ingham et al. (1974) and Latané et al. (1979), where social loafing and coordination decay originally manifested. And this is where the strongest member determines the overall performance of the team. Existing benchmarks that may fall under this category include open-ended creative brainstorming and hypothesis generation (Olson et al., 2021), divergent semantic exploration in the Divergent Association Task (DAT), and parallel synthetic instruction generation.

Conjunctive Tasks in which the outcome is established through $Y = \min_i \{y_i\}$ via weakest-link coupling). Group success requires that *all* agents or subtasks succeed without failure; an error at any intermediate stage can invalidate the entire collective output. Looking at real-world repository-level software engineering in SWE-bench and SWE-bench Verified (Jimenez et al., 2024), end-to-end multi-agent enterprise workflows in TheAgentCompany (Xu et al., 2026), interactive multi-turn operating system manipulation in OSWorld (Xie et al., 2024), and sequential tool-calling environments in WorkBench and BrowseComp-Plus

(Kim et al., 2026), we notice that they can be solved as a sequence of sub-tasks where a failure in any intermediate step causes zero task completion.

Discretionary Tasks in which the outcome is established through $Y = f_{\text{group}}(\{y_i\})$ via group discretion. In Steiner’s taxonomy, discretionary tasks grant the group complete autonomy to select its own combination rule; the task constraints themselves do not mandate how member contributions combine. Modern multi-agent meta-orchestration frameworks (e.g., router-judge architectures, Mixture-of-Agents (Wang et al., 2025), ChatEval (Chan et al., 2024), and AgentBench (Liu et al., 2024)) where a central orchestrator dynamically decides whether to average, vote, deliberate, or delegate to specialized tool-calling agents may correspond to this category.

B THEORETICAL DETAILS

B.1 THE SCALING DICHOTOMY UNDER IDEAL ASSUMPTIONS

Comparing the ideal disjunctive model (Section 2.2) and the ideal compensatory model (Section 2.3) clarifies what differs between the two task types.

Disjunctive Tasks (Selection / Voting): The candidate output space $\mathcal{Y}$ is discrete. Plurality voting and verifier selection isolate the correct modal response from dispersed errors. Under Condorcet’s jury theorem and Hoeffding’s bound, the collective failure probability decays exponentially:

$$P(\text{Disjunctive Error}) \leq \exp\left(-2N\left(p - \frac{1}{2}\right)^2\right) = \mathcal{O}(e^{-\alpha N}). \quad (5)$$

Compensatory Tasks (Statistical Aggregation): Estimates $x_i \in \mathbb{R}^+$ and signed errors $e_i = \log_{10}(x_i/g) \in \mathbb{R}$ are continuous, and the collective relies on statistical cancellation. The standard error decays as

$$\text{SE}(\bar{e}_N) = \frac{\sigma}{\sqrt{N}} = \mathcal{O}(N^{-1/2}), \quad (6)$$

and, for sub-Gaussian errors, the probability that $|\bar{e}_N|$ exceeds a fixed tolerance ϵ also decays exponentially, as $\exp(-N\epsilon^2/(2\sigma^2))$.

Under ideal independence and unbiasedness, both task types therefore enjoy exponential suppression of the probability of a large error. The difference between them lies in how they respond to correlation and bias, which we analyse next.

B.2 CORRELATED AND BIASED LLM ENSEMBLES: BREAKDOWN ACROSS TASK TYPES

The classical models in Sections 2.2 and 2.3 assume independent and unbiased members. Ensembles of samples from the same LLM violate both assumptions at the level of individual items, and the consequences differ across Steiner’s task classes.

B.2.1 DISJUNCTIVE TASKS UNDER CORRELATION AND BIAS

In discrete reasoning tasks (e.g., GSM8K, MATH-500, MMLU), agents exhibit positive pairwise correlation $\rho > 0$:

Correlated Condorcet Voting: When individual votes are exchangeable with pairwise correlation $\rho > 0$, the variance of the collective vote count is inflated by the factor $1 + (N-1)\rho$ (Boland, 1989; Ladha, 1992), which corresponds to the effective size $N_{\text{eff}}(N) = N/(1 + (N-1)\rho)$. For independent samples from one model, this correlation arises from heterogeneous item difficulty, and plurality accuracy converges to the fraction of items on which the correct answer is modal (Proposition 2.1).

Error Dispersion in Generative Search: In open-ended reasoning, the space of incorrect derivations is large. When incorrect answers are dispersed across many values while correct

derivations converge on y^* , the correct answer can remain the plurality even for $p_k < \frac{1}{2}$, which allows voting to succeed despite substantial correlation. When a model concentrates its probability on a single incorrect answer, however, additional samples reinforce that answer; our measurements show that this concentration is common, with intra-class correlations between 0.53 and 0.97.

Verifiable Truth Propagation in Deliberation: Mathematical reasoning steps can be checked, that is, correct answers are demonstrable (Laughlin & Ellis, 1986). When agents exchange rationales, an agent with a correct step can present an argument that its peers are able to verify.

B.2.2 COMPENSATORY TASKS UNDER CORRELATION AND BIAS

Continuous numerical estimation (e.g., Fermi estimation in RealFP) offers fewer structural defences against correlated errors:

Violation of Unbiasedness ($b \neq 0$): LLMs internalise miscalibrated priors about physical, demographic, and geographical quantities, producing item-level biases $\mathbb{E}[e_i | k] = b_k \neq 0$. Because geometric averaging is linear in log space, the expected team error is invariant to ensemble size:

$$\mathbb{E}[\bar{e}_N | k] = \frac{1}{N} \sum_{i=1}^N \mathbb{E}[e_i | k] = b_k \implies \text{Bias}^2(\bar{e}_N | k) = b_k^2 \quad \text{for all } N. \quad (7)$$

Averaging samples of one model cannot reduce its item-level bias.

Violation of Independence ($\rho > 0$): With equicorrelated errors, $\text{Cov}(e_i, e_j) = \rho\sigma^2 > 0$, and the variance of the average approaches a non-zero floor:

$$\text{Var}(\bar{e}_N) = \frac{\sigma^2}{N} + \frac{N-1}{N} \rho\sigma^2 \xrightarrow{N \rightarrow \infty} \rho\sigma^2. \quad (8)$$

Empirically, the intra-class correlation of signed log-errors lies between 0.70 and 0.95 (Table 3).

Limited Verifiability in Deliberation: Order-of-magnitude estimates lack verifiable intermediate steps in zero-shot dialogue. Consistent with this, revision leaves Fermi accuracy essentially unchanged (between -0.6 and $+0.7$ points).

B.3 PROOFS

Oracle accuracy. Given item k , the N outputs are independent, so the probability that none is correct is $(1 - p_k)^N$; averaging over items and applying monotone convergence gives the limit $\Pr_k(p_k > 0)$.

Plurality voting. This argument generalises Lemma 1 of Chen et al. (2024), whose difficulty indicator $d_V(x)$ equals $-\delta_k$, to arbitrary answer sets. Fix item k and a competing answer $y \neq y_k^*$. The variables $D_i = \mathbb{I}[Y_{k,i} = y_k^*] - \mathbb{I}[Y_{k,i} = y] \in [-1, 1]$ are independent with mean $\delta_{k,y} = p_k - P_k(y)$. If $\delta_{k,y} > 0$, Hoeffding’s inequality gives $\Pr(\sum_i D_i \leq 0) \leq \exp(-N\delta_{k,y}^2/2)$, and a union bound over competitors yields exponential convergence to success; if $\delta_{k,y} < 0$ for some y , the same argument shows that y_k^* loses with probability tending to one. Averaging over items yields the limit π .

Attainable voting gain. For binary outcomes, item k is decided correctly in the limit if and only if $p_k > \frac{1}{2}$, so $|\pi - \bar{p}| \leq \mathbb{E}_k[\min(p_k, 1 - p_k)] \leq 2\mathbb{E}_k[p_k(1 - p_k)]$, since $\min(p, 1 - p) \leq 2p(1 - p)$ on $[0, 1]$. Finally, $\mathbb{E}_k[p_k(1 - p_k)] = \bar{p}(1 - \bar{p}) - \text{Var}_k(p_k) = \bar{p}(1 - \bar{p})(1 - \rho)$.

Compensatory decomposition. With $e_{k,i} = b_k + \varepsilon_{k,i}$ and conditionally independent $\varepsilon_{k,i}$ of variance σ_k^2 , $\mathbb{E}[\bar{e}_{k,N}^2 | k] = b_k^2 + \sigma_k^2/N$. For equicorrelated errors with common bias b , $\text{Var}(\bar{e}_N) = \sigma^2(1 + (N - 1)\rho)/N = \sigma^2/N_{\text{eff}}(N)$, which gives the stated MSE.

C EXPERIMENTAL DETAILS: MODELS, PROMPTS, AND DATA INTEGRITY

Evaluation protocol. Models are served with vLLM with nucleus parameter 0.95 and at most 1,024 generated tokens; explicit thinking modes are disabled, and gpt-oss-20b uses low reasoning effort. Each configuration is run with three random seeds (two for qwen2.5-3b, one or two for r1-distill-qwen-14b) on the benchmark test sets: 1,319 items for GSM8K, 1,017 for GSM-Hard, 500 for MATH-500, 724 for MMLU-Hard, 1,165 for ARC-Challenge, and the 529 of 557 RealFP test items with a positive numerical answer. In Rounds 2 and 3, each agent receives its own previous response and the previous responses of all $N - 1$ teammates, each truncated to 120 tokens (answer, confidence, and the beginning of the rationale). Plurality ties are broken at random. On disjunctive tasks, ρ is the intra-class correlation of binary correctness indicators across items, macro-averaged over the five benchmarks, and $N_{\text{eff}}(N) = N/(1 + (N - 1)\rho)$. On RealFP, the item-level bias b_k is the mean signed log-error of a model on item k , ρ is the intra-class correlation of signed log-errors, and an estimate is counted as accurate if $|\bar{e}_N| \leq 1$. Confidence intervals are percentile intervals from 1,000–2,000 bootstrap resamples of items, with items resampled jointly across models and differences computed on paired items.

Models. Hugging Face identifiers, as recorded in the run manifests: HuggingFaceTB/SmolLM3-3B, Qwen/Qwen2.5-3B-Instruct, Qwen/Qwen3-4B, mistralai/Mistral-7B-Instruct-v0.3, allenai/OLMo-2-1124-7B-Instruct, Qwen/Qwen2.5-7B-Instruct, meta-llama/Llama-3.1-8B-Instruct, marin-community/marin-8b-instruct, Qwen/Qwen3-8B, microsoft/phi-4, Qwen/Qwen3-14B, deepseek-ai/DeepSeek-R1-Distill-Qwen-14B, and openai/gpt-oss-20b.

Prompts. The Round-1 prompt for GSM8K reads: “Solve this math problem step by step. Be concise. Problem: {question}. Format EXACTLY (answer and confidence FIRST, then reasoning): FINAL: {integer} CONF: {0–100} RATIONALE: {brief reasoning, max 4 lines}.” The other tasks use the same structure with task-specific answer formats. The revision prompt presents the agent’s previous response and the peer responses, each truncated to 120 tokens, and asks the agent to “update or keep” its answer in the same format. For RealFP, the prompt analysed in the main text instructs the model to decompose the quantity into a chain of factors with units, to combine them while tracking units and powers of ten, and to answer in the unit of the reference value; its revision prompt states that “your goal is NOT consensus” and asks the agent to change its estimate only to fix a concrete error in its own work. An earlier Fermi prompt, used for some runs (see below), was derived from the GSM8K template (“Give your best order-of-magnitude estimate for this Fermi problem. Reason step by step.”, with a rationale of at most four lines) and used the same revision prompt as the disjunctive tasks.

Data integrity. An audit of the generation logs identified four irregularities, all of which are corrected or excluded in the reported analyses. (i) The RealFP runs were generated with the two prompt families described above, which we identify from the agents’ outputs: in every run of the decomposition family, 14–99% of Round-2 rationales echo the diversity-preserving instruction (e.g., “I kept my estimate because . . .”), and most Round-1 outputs list numbered factors, whereas no Round-2 rationale of the earlier family does. The decomposition family covers ten of the 13 panel models (all runs of qwen2.5-3b, olmo2-7b, qwen2.5-7b, and marin-8b; the third run of mistral-7b, llama3.1-8b, qwen3-8b, phi4-14b, and qwen3-14b; and the second and third runs of smollm3-3b), and all Fermi results in the main text are restricted to these runs. gpt-oss-20b, qwen3-4b, and r1-distill-qwen-14b were run only with the earlier prompt. For the earlier family (nine models), the clipped log-error changes from 1.83 ($N = 1$) to 1.72 ($N = 5$) and 1.74 ($N = 30$), the bias share is $\beta = 0.89$, and the mean cross-model bias correlation is 0.55 (0.33 on label-clean items), so the conclusions are unchanged. (ii) One cell of the earlier family (gpt-oss-20b, RealFP, $N = 30$, third run) has 1.4 times longer outputs and half the accuracy of all other cells and is excluded. (iii) r1-distill-qwen-14b frequently reaches the 1,024-token limit: single-agent outputs are truncated on 44% of MATH-500, 31% of GSM-Hard, 29% of MMLU-Hard, and 3% of GSM8K and ARC-Challenge items, and on RealFP at $N \geq 10$; its results on these tasks should be

read as lower bounds. (iv) The original answer parser did not recognise exponents written with superscript digits and retained only the mantissa; this affected 16.0% of the Fermi answers of qwen3-14b, 5.7% of those of gpt-oss-20b, and at most 1% of those of every other model, and all Fermi answers were re-parsed before analysis.

C.1 EXAMPLE TRACES

To illustrate the mechanisms discussed in Sections 3.2 and 3.4, we show one disjunctive and one compensatory trace. To avoid selective reporting, each example was drawn at random (fixed seed) from all items that satisfy a pre-specified rule; each example is the first draw under its rule. Outputs are abbreviated with ellipses but otherwise reproduced verbatim.

Disjunctive example (llama3.1-8b, GSM8K, $N = 2$, first run). Selection rule: both agents are incorrect in Round 1 and the plurality team answer is correct in Round 3 (595 items).

Question: Carson lives in an old house where the pipes will freeze if the temperature inside the house falls below 32 degrees. Every log of wood Carson burns heats the house up by 5 degrees. If it was 45 degrees during the day, and it's 33 degrees colder during the night compared to the day, how many logs does Carson need to burn to prevent the pipes from freezing? (Reference: 4.)

Round 1, agent 1: FINAL: 6 ... "Carson needs to heat the house from 12 to 32 degrees, a total of 20 degrees. Since each log heats the house by 5 degrees, Carson needs $20 / 5 = 4$ logs to heat the house to 32 degrees. However, this calculation ..."

Round 1, agent 2: FINAL: 6 ... "To reach 32 degrees, Carson needs to heat the house by 12 degrees. ... Carson needs $12 / 5 = 2.4$ logs ..."

Round 2, both agents: FINAL: 4, with the rationale of agent 1 ("from 12 to 32 degrees, a total of 20 degrees ... $20 / 5 = 4$ logs"); unchanged in Round 3.

The two agents fail for different reasons. Agent 1 commits to an incorrect answer before reasoning and then computes the correct value in its own rationale; agent 2 makes a reasoning error and, in Round 2, adopts the reasoning of agent 1. The example therefore shows both mechanisms that can underlie the deliberation gain in Section 3.2: revising an answer in light of one's own reasoning, and adopting a peer's correct argument.

Compensatory example (qwen2.5-7b, RealFP, $N = 5$, first run). Selection rule: label-clean item, all five Round-1 estimates within one order of magnitude of each other, and a geometric mean more than one order of magnitude from the reference (115 items).

Question: If all the people in Australia joined hands and stretched themselves out in a straight line, how long would it reach? (Give your answer as a single number in meters.) (Reference: 4.57×10^7 .)

Round 1, agent 1: FINAL: 2.5e6 ... " 2.5×10^7 people $\times$ 1.5 meters/person = 3.75×10^7 meters ... 3.75×10^7 meters $\times$ 0.8 = 3×10^6 meters"

Round 1, agents 2, 3, and 5: FINAL: 2.5e6, 3e6, and 4e6.

Round 1, agent 4: FINAL: 2.5e7 ... " 2.6×10^7 people $\times$ 1.7 meters/person = 4.42×10^7 meters"

Round 3, agent 4: FINAL: 4e6 ... "since the majority of the peer estimates are around 4×10^6 meters ... my revised estimate is 2.5×10^7 people $\times$ 1.5 meters/person $/ 10 = 3.75 \times 10^6$ meters"

All agents use the same decomposition (population $\times$ arm span), and four of the five report an answer about ten times smaller than the product computed in their own rationale, so the geometric mean is one order of magnitude below the reference and inherits a shared item-level bias b_k (Section 3.4). During revision, the agent whose estimate was closest to the reference divides its own correct result by ten to match the majority, although the revision prompt explicitly states that consensus is not the goal; deliberation thus increases the error of the team.

D ADDITIONAL SCALING AND DISJUNCTIVE RESULTS

This appendix reports the exact values underlying Figure 1 (Table 4), the validation of the conditional-independence model of Proposition 2.1 (Figures 2 and 3), a temperature sweep (Table 5), and the statistics of the deliberation peak (Table 6).

Table 4: **Empirical Scaling Across Disjunctive and Compensatory Tasks.** Mean accuracy (%) across the 13-model panel, comparing Round 1 (independent answers, plurality vote or geometric mean), the Round-1 oracle success rate $\text{pass}@N$ (at least one agent correct), and Round 3 (after two revision rounds) across team sizes $N = 2$ to $N = 30$. The last column gives the largest half-width of the 95% item-bootstrap interval across team sizes in each row.

Task	Bench.	Team Size N								95% CI	
		2	3	5	7	10	15	20	25	30	half-width
Panel A: Round 1 (Independent Ensembles / Statistical Aggregation)											
Disj.	arc	82.88	83.42	83.49	83.66	83.62	83.70	83.55	83.65	83.66	±1.3
Disj.	gsm8k	34.06	34.16	34.14	34.34	34.33	34.25	34.36	34.27	34.31	±1.2
Disj.	gsmhard	18.03	18.35	18.67	18.73	18.83	18.96	18.95	19.05	19.07	±1.1
Disj.	math500	30.80	31.13	31.31	31.59	31.74	31.93	32.07	32.05	32.08	±2.1
Disj.	mmlu_hard	52.18	52.53	52.90	52.61	52.90	52.89	52.99	52.87	52.94	±2.0
Comp.	fermi*	31.15	31.81	32.89	33.12	32.82	32.91	33.33	33.14	33.35	±2.5
Comp.	fermi+	1.77	1.76	1.73	1.73	1.72	1.72	1.71	1.72	1.71	±0.07
Comp.	fermi-clean*	40.62	41.48	43.00	43.35	42.94	42.96	43.53	43.27	43.61	±2.7
Comp.	fermi-clean+	1.44	1.42	1.39	1.39	1.38	1.38	1.37	1.37	1.37	±0.06
Panel B: Round 1 Oracle Success (pass@N)											
Disj.	arc	84.82	85.83	86.61	87.12	87.46	87.90	88.14	88.26	88.43	±1.2
Disj.	gsm8k	37.79	39.93	43.31	45.47	47.83	50.32	52.17	53.47	54.44	±1.2
Disj.	gsmhard	20.32	21.62	23.22	24.29	25.47	26.92	27.95	28.41	28.87	±1.5
Disj.	math500	33.64	35.80	37.97	39.87	41.65	43.67	44.91	46.25	47.08	±2.1
Disj.	mmlu_hard	56.50	58.86	61.51	63.21	64.58	66.05	66.89	67.28	67.67	±1.9
Comp.	fermi*	37.70	41.24	46.37	48.41	51.09	53.48	55.10	56.37	57.32	±2.7
Comp.	fermi-clean*	48.39	52.80	58.72	60.96	63.86	66.67	68.47	69.48	70.80	±2.7
Panel C: Round 3 (Deliberated Teams / Multi-Turn Communication)											
Disj.	arc	84.05	84.45	84.42	84.40	84.43	84.53	84.54	84.55	84.54	±1.3
Disj.	gsm8k	60.51	61.11	61.90	61.95	61.76	61.70	61.62	61.35	60.88	±1.2
Disj.	gsmhard	32.21	32.47	32.89	33.15	33.36	33.14	33.53	33.14	33.08	±1.8
Disj.	math500	38.96	38.87	39.66	39.75	39.43	39.90	39.40	39.36	39.13	±2.3
Disj.	mmlu_hard	54.52	54.70	55.18	55.05	54.95	54.98	54.89	54.89	54.81	±2.0
Comp.	fermi*	30.57	31.76	32.55	32.72	32.70	33.04	33.21	33.16	34.01	±2.5
Comp.	fermi+	1.78	1.76	1.74	1.73	1.73	1.74	1.73	1.73	1.72	±0.07
Comp.	fermi-clean*	40.18	41.51	42.48	42.81	42.92	43.34	43.51	43.25	44.61	±2.8
Comp.	fermi-clean+	1.45	1.43	1.40	1.39	1.39	1.39	1.39	1.39	1.37	±0.06

* Accuracy within one order of magnitude (%); + mean absolute $\log_{10}$ error (clipped at ± 3). Fermi results use the decomposition-prompt runs of ten models (Appendix C); fermi-clean is restricted to the 385 items whose reference value lies within three orders of magnitude of the median model estimate (Section 3.2).

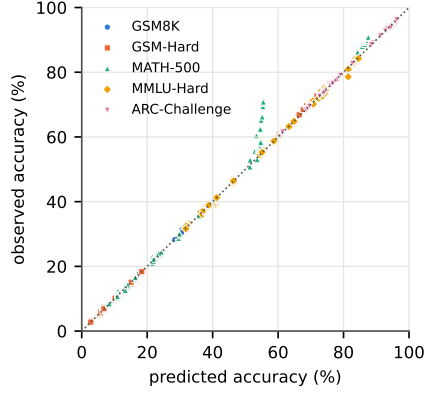


Figure 2: Observed Round-1 plurality accuracy against the accuracy predicted by resampling from the estimated item-level answer distributions, for 13 models, five tasks, and all team sizes (650 points). The dotted line is the identity. Marker shape and colour denote the task; each point is one model–task–team-size configuration.

Prediction of plurality accuracy. Figure 2 tests the disjunctive part of Proposition 2.1. For each model and task, we estimate the item-level answer distribution $\hat{P}_k$ from all Round-1 answers of the $N = 30$ teams (up to 90 samples per item across runs). For every team size N , we then draw N answers independently from $\hat{P}_k$ (200 draws per item), apply plurality voting with random tie-breaking, and average the resulting accuracy over items. The prediction thus assumes only that agents are conditionally independent given the item; it uses no information from the teams of size $N < 30$, whose answers were generated in separate runs and therefore constitute an out-of-sample test; only the $N = 30$ points are in-sample. Across 650 configurations (13 models, five tasks, and ten team sizes), the mean absolute difference between predicted and observed accuracy is 0.48 points (median 0.18; 95th percentile 1.17; Pearson $r = 0.999$). The largest deviations occur on MATH-500 for gpt-oss-20b, r1-distill-qwen-14b, and qwen3-8b, where observed accuracy exceeds the prediction: the evaluation pipeline merges mathematically equivalent answers written in different forms before voting, whereas $\hat{P}_k$ is estimated over answer strings. Excluding MATH-500, the mean absolute difference is 0.23 points and the maximum is 2.8 points.

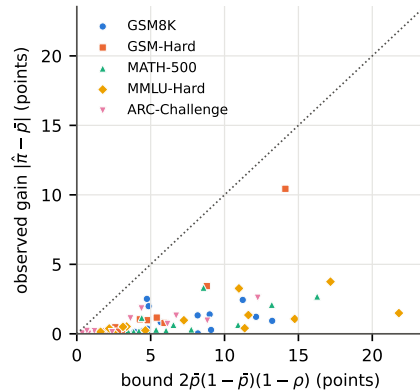


Figure 3: Validation of the voting-gain bound (Eq. 3). Observed gain of the predicted large-team plurality limit over mean single-agent accuracy, $|\hat{r} - \bar{p}|$, against the bound $2\bar{p}(1 - \bar{p})(1 - \rho)$, for the 13 panel models and five disjunctive tasks (65 points). The dotted line is the identity; every point lies below it.

Validation of the voting-gain bound. Figure 3 compares, for every panel model and disjunctive task, the observed gain $|\hat{\pi} - \bar{p}|$ with the bound of Eq. 3, computed from the mean single-agent success $\bar{p}$ and the intra-class correlation ρ of the same model and task. The bound holds in all 65 cases. It is also loose: averaged per task, the observed gain amounts to 11–21% of the bound, because the bound assumes a binary outcome, whereas incorrect answers on these tasks are often spread over several values. The bound is proven for a single dominant incorrect answer; for tasks with many answer options, Figure 3 therefore provides an empirical rather than a formal confirmation.

Sampling temperature on disjunctive tasks. Table 5 reports a temperature sweep for qwen2.5-7b with the answer-first prompts of the main experiments on GSM-Hard (400 items), MMLU-Hard (400 items), and GPQA Diamond (Rein et al., 2024) (198 items), with one run per temperature. Raising the temperature from 0.2 to 1.0 increases the dispersion of answers within a team and lowers the intra-class correlation of correctness on every task, and pass@30 rises accordingly (by 9.3, 8.3, and 14.1 points). Plurality accuracy at $N = 30$, however, remains within 2.3 points of single-agent accuracy at every temperature, and the predicted large-team limit (not shown) agrees with it to within 1 point. The additional dispersion therefore falls on incorrect answers and leaves the modal answer of most items unchanged: a lower ρ raises the bound of Eq. 3 but does not by itself realise it, and the process loss of plurality voting (pass@30 minus plurality accuracy) grows with temperature. Round-3 accuracy is also largely insensitive to temperature.

Table 5: Temperature sweep on disjunctive tasks (qwen2.5-7b, answer-first prompts, one run). Accuracy in %. “Gain” is Round-1 plurality accuracy at $N = 30$ minus single-agent accuracy, with a paired 95% item-bootstrap interval; ρ is the intra-class correlation of correctness; “Distinct” is the mean number of distinct answers among the 30 Round-1 answers to an item.

Task	T	Solo	R1, $N=30$	Gain	pass@30	ρ	Distinct	R3, $N=30$
GSM-Hard	0.2	11.0	10.0	−1.0 [−2.0, −0.2]	13.5	0.92	3.3	27.0
	0.4	11.0	10.8	−0.2 [−1.5, +1.0]	18.2	0.85	6.2	28.0
	0.7	11.0	10.8	−0.2 [−2.0, +1.5]	19.8	0.79	10.0	27.3
	1.0	10.2	11.0	+0.8 [−1.0, +2.8]	22.8	0.75	13.0	29.0
MMLU-Hard	0.2	54.5	53.2	−1.2 [−3.0, +0.2]	55.5	0.98	1.0	58.0
	0.4	55.2	53.0	−2.2 [−4.2, −0.5]	58.2	0.95	1.1	57.0
	0.7	54.8	53.2	−1.5 [−3.8, +0.5]	61.3	0.90	1.2	56.5
	1.0	54.8	53.0	−1.8 [−4.0, +0.8]	63.7	0.87	1.3	57.0
GPQA Diamond	0.2	29.3	30.8	+1.5 [+0.0, +3.5]	33.3	0.97	1.1	32.3
	0.4	30.3	30.3	+0.0 [−2.0, +2.0]	37.4	0.93	1.2	30.3
	0.7	29.8	29.3	−0.5 [−2.5, +1.5]	42.9	0.85	1.4	30.3
	1.0	30.8	29.3	−1.5 [−4.5, +1.5]	47.5	0.80	1.6	30.3

Table 6: **Anatomy of the Disjunctive Deliberation Peak and Decay.** Mean accuracy (%) across the 13-model panel: Round 1 at $N = 2$ and $N = 30$, Round 3 optimal team size N^* , peak accuracy, accuracy at $N = 30$, post-peak change $\Delta_{\text{decay}} = \text{Acc}_{N=30} - \text{Acc}_{N^*}$, and net deliberation gain at the peak $\Delta_{\text{delib}} = \text{Acc}_{N^*} - \text{Acc}_{R1, N=2}$, each with a paired 95% item-bootstrap interval. Because N^* is selected as the maximum, Δ_{decay} is biased downward.

Bench-mark	R1 $N = 2$	R1 $N = 30$	N^* Peak	R3	R3 $N = 30$	Δ_{decay}	Δ_{delib}
arc	82.88	83.66	25	84.55	84.54	−0.01 [−0.13, +0.13]	+1.66 [+1.38, +1.95]
gsm8k	34.06	34.31	7	61.95	60.88	−1.07 [−1.39, −0.73]	+27.89 [+27.01, +28.79]
gsmhard	18.03	19.07	20	33.53	33.08	−0.45 [−0.69, −0.21]	+15.50 [+14.49, +16.52]
math500	30.80	32.08	15	39.90	39.13	−0.77 [−1.22, −0.33]	+9.10 [+8.08, +10.08]
mmlu_hard	52.18	52.94	5	55.18	54.81	−0.37 [−0.73, −0.01]	+3.00 [+2.42, +3.58]

E ADDITIONAL COMPENSATORY RESULTS

This appendix reports the effect of sampling temperature (Table 8), the distribution of signed errors of single agents (Table 9), and the cross-model correlation of item-level biases (Table 10), which are discussed in Section 3.4, and it begins with a direct validation of Eq. 4 (Figure 4) and a sensitivity analysis of the clipping threshold (Table 7).

Validation of the compensatory decomposition. For each model, we estimate the item-level bias b_k and the within-item variance σ_k^2 of the clipped log-errors from the Round-1 answers of the $N = 30$ teams, and predict the mean squared error of the geometric mean as $\mathbb{E}_k[b_k^2] + \mathbb{E}_k[\sigma_k^2]/N$. We compare this prediction with the observed mean squared error of single agents and of the actual Round-1 teams of each size, which were generated in separate runs; only the $N = 30$ points are in-sample. For all ten models, the mean relative difference between predicted and observed error is 0.6% (maximum 3.0%; Pearson $r = 0.997$ over 90 out-of-sample points). The curves in Figure 4 flatten towards the bias floor $\mathbb{E}_k[b_k^2]$ within five to ten agents, which is the compensatory ceiling predicted by Proposition 2.1.

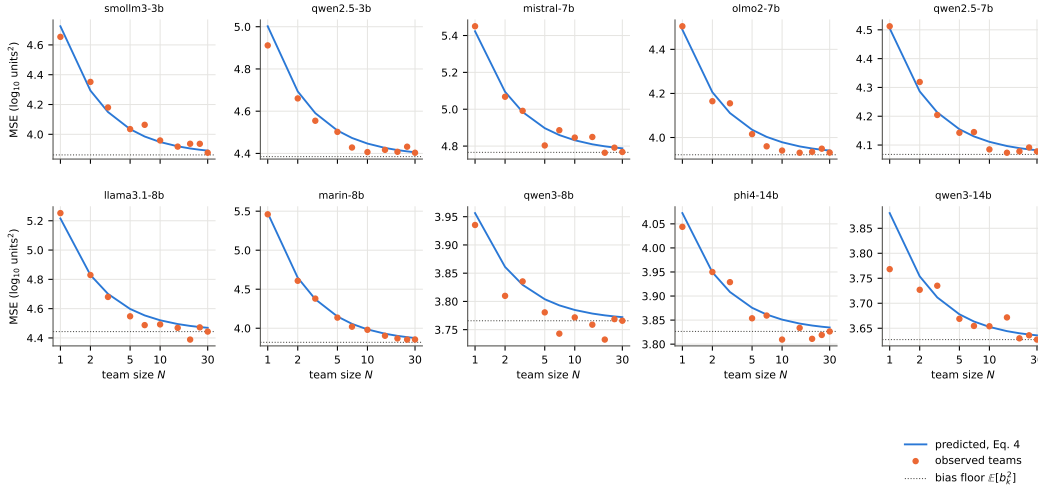


Figure 4: Validation of the compensatory decomposition (Eq. 4). For each panel model, the mean squared error of the geometric mean of clipped log-errors predicted from item-level bias and within-item variance (line) and observed for single agents and actual Round-1 teams (points), together with the bias floor $\mathbb{E}_k[b_k^2]$ (dotted). Decomposition-prompt runs. The y -axes do not start at zero, so the vertical scale exaggerates the decrease; the observed reduction in mean squared error from $N = 1$ to $N = 30$ is at most 13% for seven of the ten models and 15%, 17%, and 29% for llama3.1-8b, smollm3-3b, and marin-8b, respectively.

Sensitivity to the clipping threshold. Log-errors are clipped at ± 3 throughout, which limits the influence of implausible reference values (Section 3.2) and of occasional estimates that are wrong by tens of orders of magnitude. Table 7 repeats the main compensatory analyses with thresholds of ± 2 and ± 5 and without clipping. At every threshold, averaging reduces the error by only 4–8%, and item-level bias accounts for 86–94% of the mean squared error. Without clipping, the mean is dominated by a small number of extreme items, which raises both the bias share and the cross-model correlation; clipping therefore makes the reported ceiling conservative rather than producing it. The median absolute log-error, which requires no clipping, decreases only from 2.04 for a single agent to 1.96 at $N = 30$.

Table 7: **Sensitivity of the RealFP Results to Clipping.** Mean absolute log-error (MAE) of the geometric mean in Round 1 (single agents at $N = 1$), its relative reduction from $N = 1$ to $N = 5$ and to $N = 30$, bias share $\beta = \mathbb{E}[b_k^2]/(\mathbb{E}[b_k^2] + \mathbb{E}[\sigma_k^2])$, and mean pairwise cross-model correlation of item-level biases r , macro-averaged over the ten models with decomposition-prompt runs. The ± 3 row corresponds to the values reported in the main text.

Clipping	MAE N = 1	MAE N = 5	MAE N = 30	Red. 1→5	Red. 1→30	β	r
± 2	1.41	1.32	1.30	6.9%	7.9%	0.86	0.61
± 3	1.84	1.73	1.71	6.1%	7.1%	0.87	0.63
± 5	2.41	2.28	2.26	5.5%	6.4%	0.88	0.65
none	3.82	3.68	3.66	3.7%	4.2%	0.94	0.84

Sampling temperature. Table 8 reports Round-1 within-one-order-of-magnitude accuracy (median aggregation) on all 529 RealFP items at three temperatures, after correcting the parsing of exponents written with superscript digits (Appendix C). Accuracy is essentially invariant to temperature: all differences between temperatures lie well within the 95% intervals of about ± 4 points. Higher temperature does increase the dispersion of estimates within a team: the within-item standard deviation of log-errors grows by 30–70% from $T = 0.4$ to $T = 1.0$, and the share of unanimous teams falls from 33% to 4%. The between-item standard deviation remains several times larger, and the intra-class correlation remains between 0.83 and 0.98, so additional sampling diversity does not translate into better aggregates.

Table 8: Within-one-order-of-magnitude accuracy (%) on RealFP by temperature and team size (Round 1, median aggregation, 529 items, one to three runs per cell); $\pm$ gives the half-width of the 95% item-bootstrap interval.

Model	$T = 0.4$			$T = 0.8$			$T = 1.0$		
	$N=2$	$N=5$	$N=10$	$N=2$	$N=5$	$N=10$	$N=2$	$N=5$	$N=10$
qwen3-4b	34.3 $\pm$ 3.9	35.5 $\pm$ 3.7	35.3 $\pm$ 3.6	33.5 $\pm$ 3.6	34.7 $\pm$ 3.9	35.8 $\pm$ 3.8	33.4 $\pm$ 3.4	35.2 $\pm$ 3.8	36.2 $\pm$ 3.9
qwen3-8b	38.3 $\pm$ 4.0	38.4 $\pm$ 3.9	38.5 $\pm$ 4.1	39.4 $\pm$ 3.7	38.6 $\pm$ 3.9	38.7 $\pm$ 3.7	39.2 $\pm$ 3.9	38.8 $\pm$ 3.9	38.9 $\pm$ 4.3
qwen3-14b	38.1 $\pm$ 4.0	38.2 $\pm$ 4.2	38.3 $\pm$ 4.4	38.1 $\pm$ 3.7	37.6 $\pm$ 4.1	38.9 $\pm$ 4.2	38.9 $\pm$ 3.9	38.0 $\pm$ 4.0	38.4 $\pm$ 4.1
gpt-oss-20b	50.3 $\pm$ 3.9	55.3 $\pm$ 3.9	55.2 $\pm$ 4.0	49.9 $\pm$ 3.7	55.0 $\pm$ 3.9	55.4 $\pm$ 4.0	51.0 $\pm$ 3.8	56.1 $\pm$ 3.9	54.8 $\pm$ 4.2

Table 9: **Intra-Model Bias and Distribution of Signed Errors.** Mean signed error $b = \mathbb{E}[e_i]$ (clipped at ± 3 ; 95% item-cluster bootstrap interval), median signed error, standard deviation σ (clipped), and percentage of estimates under- and over-estimating the ground truth. Evaluated on single-agent estimates of the decomposition-prompt runs, sorted by parameter count.

Model	Obs.	$\mathbb{E}[e_i]$ (Mean b) (OOM)	Median (OOM)	σ (Std)	Under (%)	Over (%)
smollm3-3b	1,058	+0.48 [+0.32, +0.66]	+0.49	2.10	41.2	57.9
qwen2.5-3b	1,058	+0.65 [+0.47, +0.84]	+0.92	2.12	36.1	63.5
mistral-7b	522	+0.71 [+0.53, +0.89]	+1.21	2.23	35.6	63.6
olmo2-7b	1,158	+0.19 [+0.02, +0.37]	+0.28	2.11	45.1	54.5
qwen2.5-7b	1,587	+0.11 [-0.05, +0.28]	+0.02	2.12	48.9	50.1
llama3.1-8b	528	+0.62 [+0.42, +0.79]	+0.97	2.21	39.0	60.8
marin-8b	1,585	-0.20 [-0.38, -0.03]	-0.19	2.33	52.2	47.1
qwen3-8b	525	-0.04 [-0.20, +0.13]	-0.03	1.99	50.5	48.2
phi4-14b	528	+0.47 [+0.31, +0.64]	+0.35	1.96	39.8	58.7
qwen3-14b	529	+0.16 [+0.00, +0.31]	+0.08	1.94	46.1	52.0

Table 10: **Cross-Model Correlation of Item-Level Biases (r).** Pairwise Pearson correlation of item-level mean signed log-errors (clipped at ± 3) across the ten models with decomposition-prompt runs, on the items with estimates from all models, sorted by parameter count. The mean off-diagonal correlation is 0.63 (95% item-bootstrap interval [0.60, 0.67]).

	smollm3-3b	qwen2.5-3b	mistral-7b	olmo2-7b	qwen2.5-7b	llama3.1-8b	marin-8b	qwen3-8b	phi4-14b	qwen3-14b
smollm3-3b	1.00	0.69	0.55	0.66	0.64	0.64	0.65	0.62	0.58	0.64
qwen2.5-3b	0.69	1.00	0.50	0.68	0.67	0.64	0.60	0.64	0.59	0.64
mistral-7b	0.55	0.50	1.00	0.52	0.53	0.65	0.55	0.46	0.59	0.52
olmo2-7b	0.66	0.68	0.52	1.00	0.66	0.61	0.66	0.66	0.61	0.66
qwen2.5-7b	0.64	0.67	0.53	0.66	1.00	0.69	0.61	0.66	0.71	0.67
llama3.1-8b	0.64	0.64	0.65	0.61	0.69	1.00	0.61	0.60	0.68	0.67
marin-8b	0.65	0.60	0.55	0.66	0.61	0.61	1.00	0.63	0.61	0.65
qwen3-8b	0.62	0.64	0.46	0.66	0.66	0.60	0.63	1.00	0.73	0.77
phi4-14b	0.58	0.59	0.59	0.61	0.71	0.68	0.61	0.73	1.00	0.72
qwen3-14b	0.64	0.64	0.52	0.66	0.67	0.67	0.65	0.77	0.72	1.00

F DE-SATURATING COMPENSATORY LLMs: INTERVENTIONS AND SPECIALIST MODELS

Having established that homogeneous LLM teams do not overcome the compensatory ceiling on their own, we explore structured interventions intended to decorrelate errors ($\rho \rightarrow 0$) or to reduce systematic bias ($b^2 \rightarrow 0$).

F.1 THE HETEROGENEOUS INTERVENTION BATTERY

We evaluate interventions on a heterogeneous team of five 7B–8B models (qwen2.5-7b, llama3.1-8b, mistral-7b, olmo2-7b, and marin-8b; one run, $T = 0.4$, 529 items):

1. **Heterogeneous Baseline:** the five models without intervention.
2. **Prefix Anchoring:** each agent receives a different decomposition strategy (e.g., top-down, bottom-up, rate $\times$ time) as a prompt prefix.

Table 11: Heterogeneous-team interventions on RealFP (Round 1, median aggregation). Accuracy within one order of magnitude (%) and mean absolute $\log_{10}$ error (unclipped, and therefore not comparable with the clipped values of Section 3.4; see Table 7) at $N = 5$, with 95% item-bootstrap intervals at $N = 5$. Where fewer than 529 items were completed, n is given.

Intervention	Acc. $N=2$	Acc. $N=3$	Acc. $N=5$	Error $N=5$
Baseline (no intervention)	28.9	31.4	33.1 [29.3, 36.9]	3.41 [2.91, 3.96]
Decomposition anchors	28.0	30.4	33.1 [29.3, 37.1]	3.44 [2.98, 3.96]
Unit systems + logcalc	18.0	20.2	25.0 [21.4, 28.9]	4.52 [3.95, 5.15]
Logcalc	18.5	24.4	28.9 [22.2, 35.6] ($n = 180$)	5.03 [3.81, 6.35]
Grounding + logcalc	21.0	19.7	21.1 [10.5, 31.6] ($n = 57$)	4.35 [2.96, 6.42]
Distinct scaffolds	30.2	33.6	36.3 [32.1, 40.6]	3.31 [2.83, 3.88]
Anchors + logcalc	18.9	19.7	26.1 [22.5, 29.9]	4.51 [3.97, 5.13]

3. **Structured Scaffolding:** each agent follows a different complete estimation template.
4. **Log-Calculator Offloading (logcalc):** agents list estimated factors, and the answer is computed from them in log space.
5. **Unit System Change-of-Basis:** agents reason in different unit systems, combined with logcalc.
6. **Constant Grounding:** a table of reference constants is inserted into the prompt, combined with logcalc.

From the results reported in Table 11 it can be seen that Structured scaffolding yields the highest accuracy at $N = 5$ (36.3%, 95% CI [32.1, 40.6], compared with 33.1%, [29.3, 36.9], for the baseline) and the lowest mean log-error (3.31, unclipped), although its interval overlaps with that of the baseline. Every logcalc variant reduces accuracy relative to the baseline. The logcalc and grounding conditions at $N = 5$ were completed for only 180 and 57 items, respectively, and are not comparable with the other cells. Among homogeneous runs, logcalc also reduced accuracy for gpt-oss-20b (from 52.0% to 48.0% for a single agent), and decomposition anchors had no discernible effect.

F.2 EVALUATION OF A SPECIALIST MODEL

We also compared Ornith-1.5-9B, a publicly available model fine-tuned from Qwen3.5-9B with reinforcement learning for agentic coding, with its base model Qwen3.5-9B on RealFP and on a Science Olympiad estimation set (SciOly, 500 items), using Program-of-Thought (PoT) and decomposition-scaffold prompting. The control runs with direct and chain-of-thought prompting used a smaller token budget and were truncated on about half of the items, so we report only the paired comparison on items that both models answered with the same 16k-token budget. The fine-tuned model is not more accurate on RealFP (+0.4 points for PoT, 95% CI [-3.6, 4.4]; +0.4 for the scaffold, [-3.9, 4.9]) or with PoT on SciOly (+0.7, [-2.6, 3.7]), and is more accurate only with the scaffold on SciOly (+5.7, [0.7, 10.4]). Program-of-Thought prompting yields the highest accuracy for both models (about 50% within one order of magnitude on RealFP and 76% on SciOly).